

Predicting Human Intervention in Robotic Manufacturing Environments

Project Description

Robotic systems are dominating the manufacturing environments tasks such as welding, painting, assembly, and inspection. However, the performance of these robots can vary depending on the task type, sensor input, environmental conditions, and system load. The project will focus on predicting task reliability and likelihood of human intervention using operational, sensor, and environmental data. The primary goal will be to use the ML model to identify patterns that influence task reliability, efficiency, and likelihood of human intervention, which can help improve system performance, reduce downtime, and limit the need for human oversight. The dataset used represents performance of robots in real world environments and can be used to examine how they perform in various settings. It also helps examine how sensor data and energy usage relate to task issues such as defects and the need for human involvement.

Dataset

Dataset Name: Robotic Operations Performance Dataset

Source: Kaggle

Direct Link: <https://www.kaggle.com/datasets/ziya07/robotic-operations-performance-dataset>

Description: The dataset includes information on the type of tasks being performed by robots (Welding, Painting, Assembly, Inspection), along with associated sensor data and performance metrics. It will demonstrate how robots perform in dynamic environments and whether human intervention or defect detection occurs during tasks.

Input Features and Output Target

Input Features (Independent Variables):

- Task_Type (Welding, Painting, Assembly, Inspection): Type of task robot was performing, such as welding, painting, or assembly
- Sensor_Type: Type of sensor used to monitor robot's environment (LIDAR, Camera, Thermal)
- Sensor_Data: Data gathered by sensor, includes information such as obstacle detection, temperature readings, or object recognition accuracy
- Environmental_Status: Indicates whether robots operating environment is stable or unstable
- Energy_Consumption (kWh): Amount of energy consumed by robot while performing the task

- Obstacle_Detected (Yes/No): Indicates whether obstacle was detected during specific task
- Defect_Detected (Yes/No): Indicates whether an obstacle was detected during task

Output Target (Dependent Variable):

- Human_Intervention_Needed (Yes/No): The model's goal is to predict whether a robotic task will require human intervention based on task conditions, sensor input, and environmental factors.

Type of Machine Learning Problem

Supervised Learning

- Classification

Justification

- This is a supervised learning problem because the dataset contains labeled features and a binary output variable (Human_Intervention_Needed). The model will classify whether human intervention was required based on specific tasks being performed, sensor data, and environmental conditions.